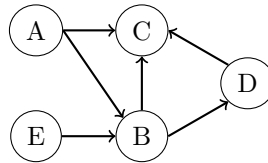


Worksheet 3

Exercise 1: D-Separation

Have a look at the following causal graphical model from the worksheet 2.

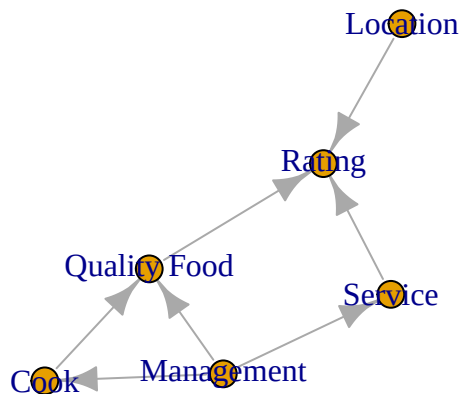


Check the following d-Separations.

1. Are E and D d-separated by B?
2. Are E and C d-separated by A?

Exercise 2: Rating of a restaurant

The following causal graph is showing factors that influence the overall rating of a restaurant.



- a) How does this graph look like if we plan an intervention on quality of food?
- b) List all possible adjustment set given by the backdoor criterion to estimate the causal effect of the cook on the rating?

Exercise 3: Exam Score

Imagine we want to analyze the effect of study hours on exam scores. However, a student's prior knowledge influences both the amount of time spent studying and the exam performance. The data of this study are available in the data.frame `exam`, which is stored in the file `exam.rda`.

- a) Draw the causal diagram.

- b) What is the probability of getting a good grade on the exam?
- c) Calculate the conditional probability of a high score given that a student studies much, i.e. $P(\text{score} = \text{high} \mid \text{study} = \text{much})$, and the conditional probability of a high score given that a student studies little, i.e. $P(\text{score} = \text{high} \mid \text{study} = \text{little})$.
- d) Calculate the probability of a high exam score under the hypothetical intervention that all students have studied much, i.e. $P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{much}))$.
- e) Calculate the probability of a high exam score under the hypothetical intervention that all students have studied little, i.e. $P(\text{score} = \text{high} \mid \text{do}(\text{study} = \text{little}))$.
- f) What can you say about the effect of learning on the exam grade, taking into account the calculations from the previous subtasks?